

AI DRIVEN ESTIMATION AUTOMATION

Overview Report

1. Construction Estimation Scope

Based on discussions with the Estimation Team, construction estimation covers the following major material and work categories. Concrete, Steel and Earthwork are identified as major aspects.

SL No	Material / Work Category	Importance	Current AI Coverage / Scope
1	Concrete	High Priority	Beam & Slab currently developed
2	Steel	High Priority	Beam currently developed
3	Blockwork	Standard Scope	Wall / block quantities
4	Tiles & Granite	Standard Scope	Floor / wall finishes
5	Plastering	Standard Scope	Internal / external surfaces
6	Painting	Standard Scope	Paintable surfaces
7	Waterproofing	Standard Scope	Wet areas / terraces
8	Doors & Windows	Standard Scope	Openings / glazing
9	Earthwork	Standard Scope	Excavation / filling quantities

2. Major Structural Components

The structural estimation framework is organized around six major components. The current development covers Concrete for Beam and Slab and Steel for Beam.

Component	Concrete	Steel	Current Position
Beam	Developed	Developed	Current development focus
Slab	Developed	Future	Major structural component
Column	Future	Future	Structural member
Foundation	Future	Future	Foundation estimation
Shear Wall	Future	Future	Vertical structural wall
Retaining Wall	Future	Future	Earth-support structure

3. Construction Estimation Involves

INPUT → INTERPRET & MEASURE → CALCULATE → VALIDATE → OUTPUT

- **INPUT:** Drawings, schedules, notes and revisions.
- **INTERPRET & MEASURE:** Geometry, dimensions, areas, lengths and counts.
- **CALCULATE:** Engineering rules, formulas and project practices.
- **VALIDATE:** Quantity checks, anomalies and estimator review.
- **OUTPUT:** Quantity take-off, BOQ-ready data and Excel outputs.

AI reduces repetitive interpretation effort, while engineering logic remains central to quantity calculation.

4. Current AI Development — What Has Been Built

CONCRETE — BEAM & SLAB	STEEL — BEAM
• Beam concrete	• Reinforcement drawing understanding
• Beam shuttering	• Beam matching / ownership
• Slab concrete	• Reinforcement parsing
• Slab shuttering	• Steel quantity + BBS
• Deterministic calculation	• Hybrid CAD + Vision
• LLM assistance for ambiguous inputs	• Deterministic engineering authority

Common foundation: reusable drawing intelligence → material/component-specific estimation engines

5. Concrete Beam & Slab Estimation

The Concrete Beam & Slab Estimation model processes structural DXF framing drawings, identifies structural members, performs engineering calculations and produces estimator-oriented outputs.

DRAWING → ENGINEERING → OUTPUT

DRAWING	ENGINEERING	OUTPUT
DXF framing drawing Member detection Geometry / dimensions	Concrete volume Shuttering area Rules & calculations	Excel quantity report Review sheet Flagged items

- Standard drawings can be processed deterministically.
- LLM assistance is used only when information such as slab thickness, tags, beam dimensions or bounded regions is missing / ambiguous.
- Estimation Team validation demonstrated meaningful productivity improvement; quantity accuracy still requires further improvement.

Concrete Beam & Slab Estimation Model Link: <http://13.201.15.89>

6. Concrete POC — Estimation Team Validation: Accuracy & Time Saving

Metric	Result
Slab Concrete	69%
Slab Shuttering	73%
Beam Concrete	78%
Beam Shuttering	81%
Slab Productivity	29% time saving
Beam Productivity	67% time saving

Estimation Team conclusion: processing effort is reduced, but current quantities still require accuracy improvement.

7. Steel Beam Estimation — Hybrid AI Architecture

The Steel Beam model uses Hybrid AI architecture that combines deterministic CAD / engineering information with AI Vision for semantic reinforcement interpretation. Final quantity calculation remains under deterministic engineering control.

DETERMINISTIC CAD / ENGINEERING	AI VISION	HYBRID RESOLUTION
Geometry Beam model Spans / dimensions Cover Engineering rules	Reinforcement groups Bar count / diameter Top / Bottom / Main / Extra Support scope Visual stirrups	Combine semantic interpretation with deterministic facts and project rules before quantity computation.

VISION: “What reinforcement exists?” → Hybrid Resolution → ENGINEERING: “How is it quantified?”

8. Steel Beam Model — Engineering Intelligence Stack

Capability	Coverage
Drawing Architecture	Identity • drawing sets • beam index • versioning
Geometry Intelligence	Regions • sketches • text • leaders • blocks
Detail & Match	Multi-view recognition • context • candidate ranking • confidence
Ownership	Text / leader / bar / stirrup / shared ownership
Reinforcement Parsing	Longitudinal bars • stirrups • side-face • laps • hooks • curtailment
Quantity & Output	Cut lengths • unit weights • steel quantity • BBS • Excel

9. End-to-End Steel Estimation Workflow

1. Upload drawing package
2. Understand drawing
3. Extract geometry & context
4. Identify components / material regions
5. Route to estimation engine
6. AI / Vision where required
7. Hybrid resolution
8. Deterministic quantity calculation
9. Validation & review
10. BOQ / BBS / Excel

10. Steel Beam — Internal Accuracy Testing

The following results represent current internal model testing. They are not final acceptance results and remain subject to Estimation Team validation.

Diameter (mm)	Automated (MT)	Estimated (MT)	Difference	Error Scale	% Error	% Accuracy
Y8	0.55	0.43	0.12	Overestimate	27.57%	72.43%
Y10	0.91	1.18	-0.27	Underestimate	-23.12%	76.88%
Y12	2.31	2.78	-0.47	Underestimate	-16.98%	83.02%
Y16	3.51	2.57	0.94	Overestimate	36.49%	63.51%
Y20	3.00	2.33	0.67	Overestimate	28.54%	71.46%
Y25	1.31	1.12	0.19	Overestimate	17.25%	82.75%
TOTAL	11.58	10.41	1.17	Overestimate	11.24%	88.76%

Internal testing result: 70–80% overall accuracy • Final performance subject to Estimation Team validation.

Note: the diameter-wise table shows an aggregate of 88.76% accuracy based on the listed automated and estimated steel quantities. The presentation summarizes the broader current internal testing outcome as 70–80% overall accuracy. Both are internal testing indicators until validated by the Estimation Team.

Steel Beam Estimation Model Link: <http://13.127.104.99/>

11. Complete AI Estimation Platform — Overall Direction

The current Concrete and Steel work is positioned as the foundation for a broader multi-material and multi-component AI Estimation platform.

Platform Layer	Purpose
1. Input	DXF / CAD • PDF • Image • schedules
2. Drawing Intelligence	Parsing • OCR / text • symbols • regions • spatial context
3. Engineering Geometry	Dimensions • spans • areas • lengths • member geometry
4. Material / Component Engines	Concrete • Steel • Earthwork • Blockwork • Finishes • Openings
5. Quantity Engine	Engineering rules • formulas • units • estimator practices
6. Validation	Confidence • anomalies • reconciliation • human review
7. Output	BOQ • BBS • Excel • summaries • audit trail

12. Expected Business Value

- Reduce repetitive manual drawing interpretation and quantity take-off effort.
- Improve estimation productivity while retaining engineering review where needed.

- Standardize quantity calculation logic and estimator practices.
- Improve traceability from estimated quantity to drawing evidence and engineering rules.
- Create a scalable foundation for multiple materials and structural components.
- Support faster handling of drawing revisions and repeated project estimation.
- Build structured data and validation feedback for continuous AI improvement.

13. Core Design Principles

- Engineering correctness takes priority over model complexity.
- AI interprets difficult drawing semantics; deterministic engineering logic calculates quantities wherever possible.
- AI outputs are reconciled with geometry, engineering rules and estimator practices.
- Low-confidence or anomalous cases are surfaced for review rather than silently accepted.
- The architecture is modular and reusable across materials and structural components.
- Validation of varied and unseen drawings is essential before production reliance.

14. Model Architecture



The long-term objective is a unified AI-assisted construction estimation platform covering the major material categories and structural components identified with the Estimation Team. The current Concrete Beam/Slab and Steel Beam work provides the foundation for this broader estimation automation capability.